Lab 3: Component Modelling & Architectural Pattern Selection

Customizable Subscription Box Scheduler

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Problem Statement #35: Customizable Subscription Box Scheduler

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Contents

1. Component Diagram (draw.io) — modules, interfaces & dependencies
2. Written Architecture Justification — selected pattern & rationale

Selected Architecture

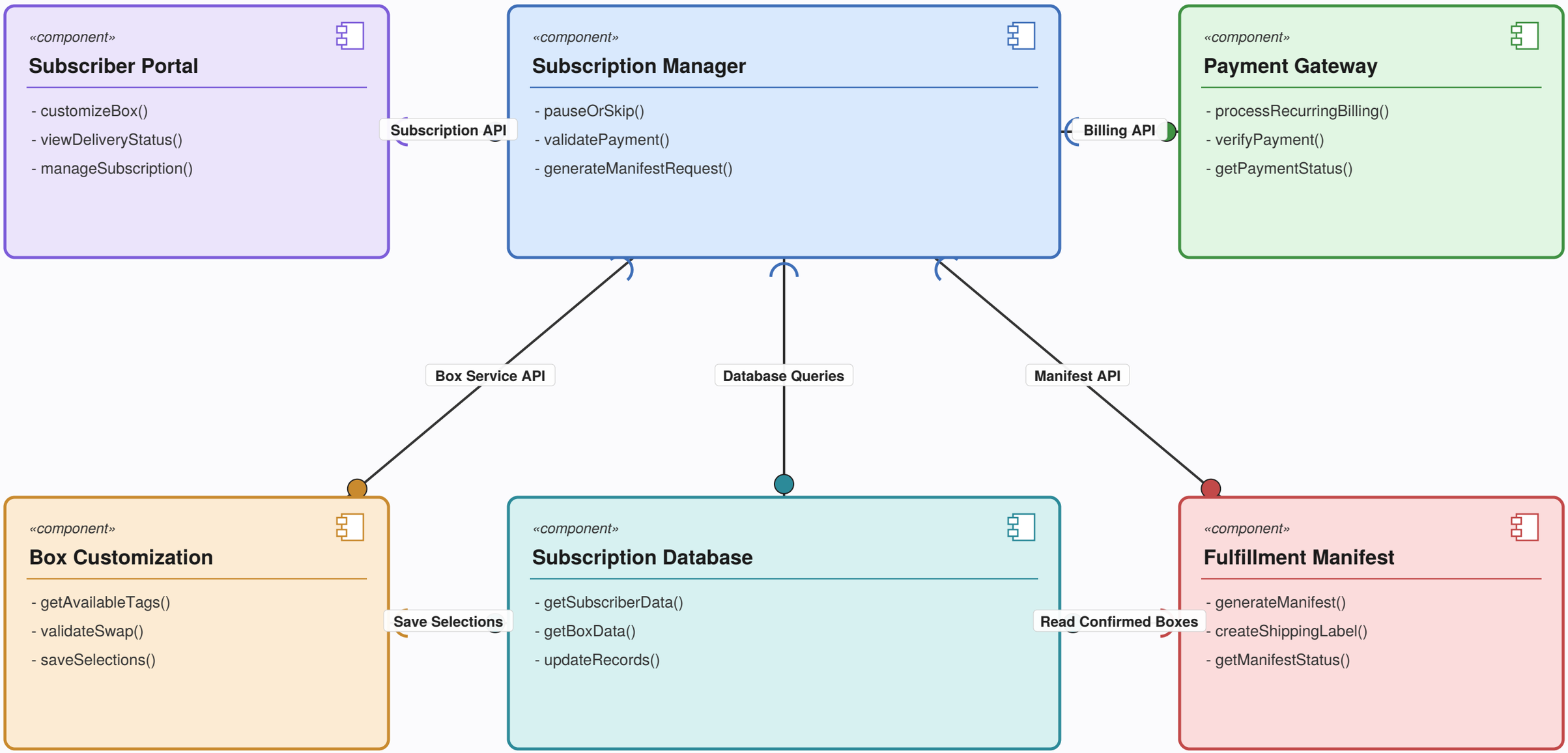
Layered Architecture (Presentation / Business / Data & External Services)

Component diagram authored natively in draw.io (diagrams.net); editable .drawio source included in the GitHub submission.

Customizable Subscription Box Scheduler

Component Diagram

(Layered Architecture)



Interfaces Used (7 total, min. 4 required)

1. Subscriber Portal -- Subscription Manager (customization, pause/skip, tracking)
2. Subscription Manager -- Box Customization (available tags, validate swap)
3. Subscription Manager -- Payment Gateway (recurring billing, payment status)
4. Subscription Manager -- Fulfillment Manifest (manifest + shipping labels)
5. Subscription Manager -- Subscription Database (subscriber / box data)
6. Box Customization -- Subscription Database (save selections)
7. Fulfillment Manifest -- Subscription Database (read confirmed boxes)

Legend

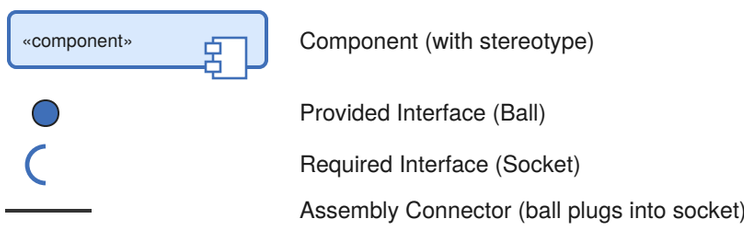


Diagram source: PES1UG24CS163_Lab3_Component_Diagram.drawio (open in app.diagrams.net / draw.io desktop to edit). Traced from Lab 1 requirements and use-case flow.

Lab 3: Architectural Pattern Selection and Component Modelling

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Architecture Selection

We chose Layered Architecture for the Customizable Subscription Box Scheduler.

Reason 1

The system has clear responsibilities: the Subscriber Portal handles user interactions, the business components manage customization, pause/skip rules and fulfillment generation, and the data layer stores subscriber, subscription and box information. This separation makes the system easier to maintain and test.

Reason 2

The architecture suits the application's centralized subscription workflow. The 48-hour customization rule, preference-tag filtering, recurring billing, and fulfillment manifest generation can be coordinated through well-defined business services without the operational complexity of deploying many independent microservices.

Security Advantage

The Payment Gateway is isolated behind a dedicated payment interface. Subscriber personal and payment-related data can be restricted to authorized services, while TLS 1.2+ protects communication and AES-256 protects stored data, matching the security requirement from Lab 1.

Performance Benefit

Frequently accessed menu, preference and subscription data can be retrieved from the data layer with minimal overhead. The Fulfillment Manifest component can generate shipping labels in a dedicated process, supporting the requirement to export labels for 10,000 monthly boxes in under one minute.

Components Identified

Subscriber Portal, Subscription Manager, Box Customization, Fulfillment Manifest, Subscription Database, and Payment Gateway.

Interfaces Shown

Subscription API, Box Service API, Manifest API, Database Queries, Save Selections, Read Confirmed Boxes, and Billing API (shown as UML ball-and-socket assembly connectors in draw.io; see component_diagram.drawio).